Local curvature of the nuisance-profile score

Incidental initial states: a local bias calculation

Increasing the number of record-specific initial states raises an incidental-parameter issue [1, 2]. Its presence alone does not establish a nonzero bias for a particular Gaussian mean model. Bias reduction for nonlinear panels has an extensive literature [3, 4]; the calculation below concerns a fixed-grid Gaussian ODE mean with known covariance, and uses a small-noise expansion rather than a large within-record sample limit.

For one record whiten the observations by the known base covariance: $y = \mu(\theta_0, u) + \epsilon z$, $z \sim N(0, I)$. At the true parameters write $S = \partial_\theta \mu$, $Z = \partial_u \mu$, $N = Z^\top Z$ and $P = I - ZN^{-1}Z^\top$. Assume the mean map is three times continuously differentiable, N is nonsingular, and the interior local nuisance minimizer admits a second-order expansion with integrable, uniform remainders over the allowed initial states. Define

$$h(u) = \sum_{i,j} (N^{-1})_{ij} \partial_{u_i u_j} \mu.$$

Local profile-score expansion. For the unscaled least-squares profile equation $\psi(\theta) = \partial_\theta \mu(\theta, \hat{u}(\theta))^\top \{y - \mu(\theta, \hat{u}(\theta))\}$,

$$E\psi(\theta_0) = -\tfrac{1}{2}\epsilon^2 S^\top P h(u) + o(\epsilon^2).$$

The log-likelihood score divides this expression by ϵ^2; the order statement above concerns the unscaled equation. Suppose the average local information $\overline{I}_e = \lim_n n^{-1} \sum_r S_r^\top P_r S_r$ is positive definite, the population profile equation has a locally unique root, and its uniform expansion is valid. That root has

$$\theta_\epsilon^\dagger - \theta_0 = -\tfrac{1}{2}\epsilon^2 \overline{I}_e^{-1} \lim_n n^{-1} \sum_r S_r^\top P_r h(u_r) + o(\epsilon^2).$$

Proof. Write $\hat{u} = u + \epsilon a + \epsilon^2 b + o_{L^2}(\epsilon^2)$. The nuisance normal equation gives $a = N^{-1}Z^\top z$. The whitened residual is $\epsilon P z - \epsilon^2 \{Zb + \frac{1}{2}\mu_{uu}[a, a]\} + o_{L^1}(\epsilon^2)$. Gaussian orthogonality makes a independent of Pz, and $E(aa^\top) = N^{-1}$. Expanding the nuisance normal equation to second order gives $Eb = -\frac{1}{2}N^{-1}Z^\top h$. The mixed coefficient-sensitivity term $E\{(\mu_{\theta u} a)^\top P z\}$ is zero. Substitution into the coefficient equation therefore gives its stated mean. Its population derivative at zero noise is $-S^\top P S$, by differentiating the nuisance normal equation. The local implicit-function expansion after averaging records gives the displayed root displacement. □

A nonzero curvature coefficient predicts a fixed-noise displacement that remains when the number of records increases. A consistency claim for the computed local profile estimator additionally requires a uniform law of large numbers and identification of this population root. Other nuisance branches, boundary optima and rare large-noise fits fall outside this local expansion. Linear mean models can have zero curvature term. The experiment tests the size and direction of the displacement rather than assuming a likelihood failure.

For the prescribed Van der Pol initial-state distribution, the deterministic Kaggle calculation gives the coefficient of ϵ^2 as approximately $(0.0024214, -0.0037071, 0.0349238, -0.0152314)$. Eight- and twelve-node Gauss–Legendre rules per initial-state coordinate agree to relative 1.8×10^{-9}; finite-difference checks of diagonal second initial sensitivities have relative errors below 3.3×10^{-6}. At $n = 800$, the predicted relative bias divided by the nuisance-adjusted norm standard error is 0.0803, 0.1605 and 0.3210 at 5%, 10% and 20% noise. These are local predictions, separate from the Monte Carlo pilot and the main comparison.

References

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